

ULPF

Universal Log Pre-processing Framework

“Different Logs. One Standard. Trusted Everywhere.”

SOFTWARE REQUIREMENTS SPECIFICATION

Functional, data and non-functional requirements baseline

Team & Role Allocation

Member	Role	Primary Ownership in This Document
Divyansh Rajat	Team Lead / System Architect	Overall architecture, scope, requirement baseline, traceability
Ashish Bhardwaj	Log Processing Engineer	Ingestion, format detection, template mining, parser factory
Sakshi	Normalization & Schema Engineer	Unified event model, OCSF/ECS mapping, schema registry, traceability
Parampreet Kaur	Backend & Integration Engineer	APIs, persistence, deployment, downstream connectors
Aryan Jeet	Frontend Engineer	Review console, dashboards, visualization requirements
Saksham	Testing, Documentation & Demo	Test strategy, benchmark methodology, acceptance evidence

Reference Documents

Ref.	Document	Source
R1	SIH 2026 Problem Statement SIH26156 — Universal Log Pre-processing Framework	Smart India Hackathon 2026 portal
R2	Drain3 — online log template mining library	logpai / IBM Drain3, open source
R3	OCSF — Open Cybersecurity Schema Framework	ocsf.io schema specification
R4	ECS — Elastic Common Schema field reference	Elastic documentation
R5	Splunk Common Information Model (CIM) documentation	Splunk Docs
R6	IBM QRadar DSM and DSM Editor documentation	IBM Documentation
R7	RFC 5424 — The Syslog Protocol	IETF
R8	ArcSight Common Event Format (CEF) and IBM LEEF specifications	Vendor specifications

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for ULPF, the Universal Log Pre-processing Framework, developed by Team S.W.O.R.D. for Smart India Hackathon 2026 problem statement SIH26156.

The document is intended for the development team as the authoritative requirement baseline, for evaluators assessing completeness and feasibility, and for any future implementer extending or integrating the system. It specifies what the system shall do; the accompanying Technical Requirements Document specifies how.

1.2 Scope of the Software

ULPF is a self-hosted preprocessing framework that receives log events from heterogeneous sources, determines or discovers their structure, extracts and normalizes their fields into a common extensible event model, preserves the complete original event, maintains bidirectional traceability between representations, and delivers standardised events to downstream security and analytics platforms.

ULPF shall not perform threat detection, correlation, alerting or case management. It shall not act as a long-term archival platform, an endpoint collection agent, or a multi-tenant hosted service. It is a layer that sits before a SIEM, data lake or machine-learning pipeline, and is designed to operate without any dependency on external network connectivity.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
CEF	Common Event Format — a text log format widely used for security events
Core schema	The stable, versioned set of fields in the ULPF unified event model
Dead letter	An event that failed processing, captured with diagnostics rather than discarded
Drain3	An open-source online log template mining library based on the Drain algorithm
ECS	Elastic Common Schema — a common set of event field names for normalization and analytics
Extension	A namespaced container holding source-specific fields that do not map to the core schema
LEEF	Log Event Extended Format — IBM's event format consumed by QRadar
Mapping	The persisted association between a source's extracted fields and target schema fields
Normalized event	The output event conforming to the unified event model
OCSF	Open Cybersecurity Schema Framework — a vendor-agnostic extensible security event schema
Parsed event	The intermediate representation after field extraction but before schema mapping
Provenance record	The stored derivation of a normalized field from its source field
Raw event	The original event exactly as received, before any transformation
Raw Event Vault	The immutable store holding raw events with integrity digests
Source	A registered, named origin of log events with associated configuration
Template	A discovered structural pattern with constant and variable positions
Trace ID	The identifier binding the raw, parsed and normalized forms of a single event
Unified event model	ULPF's common event representation: core schema plus extensions plus provenance

1.4 References

See the Reference Documents table in Document Control. Requirements derived directly from the problem statement are tagged PS in the Source column of the requirement tables.

1.5 Document Conventions

- **Shall** denotes a mandatory requirement.
- **Should** denotes a recommended requirement whose omission requires justification.
- **May** denotes an optional capability.
- **Priority** is expressed as High (required for the prototype), Medium (required for a complete product), or Low (desirable).
- **Requirement IDs** are stable. FR-n.m identifies functional requirements; NFR-x.n identifies non-functional requirements; DR-n identifies data requirements; IR-n identifies interface requirements.

2. Overall Description

2.1 Product Perspective

ULPF is a new, self-contained system positioned as middleware. Its upstream boundary is the set of log-producing devices, applications and services in the deployed environment. Its downstream boundary is the set of consuming platforms: SIEMs, data lakes, machine-learning pipelines, dashboards and REST clients.

The system is decomposed into eight logical subsystems, each addressed by a group of functional requirements in Section 4:

#	Subsystem	Responsibility	FR group
S1	Ingestion Gateway	Accept events over file, TCP, UDP, HTTP and streaming transports; bind each to a registered source	FR-1
S2	Format Detection Engine	Classify incoming structure; route known formats to parsers and unknown formats to discovery	FR-2
S3	Adaptive Parser Factory	Discover templates, extract and type fields, propose semantic mappings, score confidence	FR-3
S4	Normalization Engine	Map parsed fields into the unified event model; normalize values; place unmapped fields in extensions	FR-4
S5	Raw Event Vault	Store the complete original event immutably with an integrity digest and reception metadata	FR-5
S6	Traceability Layer	Maintain trace identifiers and field-level provenance; serve bidirectional lookup	FR-6
S7	Schema Registry	Hold versioned core schemas, mapping profiles and vendor extension namespaces	FR-7
S8	Review Console & Delivery	Human-in-the-loop mapping review, dashboards, query API and downstream output	FR-8, FR-9

2.2 Product Functions — Summary

At the highest level the system performs the following functions in sequence for every event:

- Receive the event and record it verbatim with reception metadata and an integrity digest.
- Allocate a trace identifier binding all downstream representations of that event.
- Determine the event's format; if the format or source is unknown, discover its structure.
- Extract fields and infer their data types.
- Map extracted fields to the unified event model, using deterministic rules where they apply and semantic mapping where they do not.
- Score the confidence of each mapping; hold uncertain mappings for human review rather than applying them.
- Normalize values, place unmapped fields into namespaced extensions, and emit the standardised event.
- Record field-level provenance and make both forward and reverse lookup available.
- Deliver the standardised event to configured downstream consumers.

2.3 User Classes and Characteristics

User class	Characteristics	Privileges required
Security Analyst	Consumes normalized events; occasional trace-back during investigation; medium technical depth	Read events, read provenance
Detection / Content Engineer	Onboards sources and adjudicates mapping proposals; high technical depth in log semantics	All analyst rights plus approve mappings

User class	Characteristics	Privileges required
Platform Administrator	Configures sources, transports, thresholds and schema versions; operates the deployment	Full configuration and operational rights
Compliance / Audit Officer	Retrieves raw events and provenance as evidence; low technical depth	Read raw events, read provenance, read audit trail
Downstream System	Machine consumer retrieving normalized events via API, export or forwarding	Scoped service credential, read-only on events

2.4 Operating Environment

Aspect	Requirement
Host OS	Linux x86-64 (Ubuntu 22.04 LTS or equivalent); Windows via container runtime
Runtime	Container runtime with Compose support
Network	Operates fully within an isolated network; no outbound Internet access required at install or runtime
Minimum prototype hardware	4 vCPU, 8 GB RAM, 50 GB storage
Recommended hardware	8 vCPU, 16 GB RAM, SSD storage sized to the retention window
Accelerators	None required; semantic mapping shall operate on CPU
Client	Modern evergreen browser for the review console

2.5 Design and Implementation Constraints

- C1 — No component on the processing path shall require outbound Internet connectivity, a cloud API or an external inference service.
- C2 — All third-party components shall be open-source under licences permitting redistribution in an offline bundle.
- C3 — The system shall be deployable as containers orchestrated by a single Compose definition.
- C4 — The core event schema shall align with an established open standard (OCSF) rather than a proprietary schema.
- C5 — Model weights required for semantic mapping shall be bundled at build time, not downloaded at runtime.
- C6 — Raw event storage shall be write-once; no processing path may modify a stored raw event.
- C7 — All configuration shall be file-based and version-controllable; no configuration shall exist only in a database or UI state.

2.6 Assumptions and Dependencies

- A1 — Log events are text-based or textually representable; encrypted and binary proprietary payloads are out of scope.
- A2 — A sufficient sample volume from a new source is available for template mining to converge.
- A3 — Reviewers with domain knowledge are available to adjudicate low-confidence mappings.
- A4 — The deployment host clock is synchronised; ULPF records its own receipt time independently of source timestamps.
- D1 — The system depends on an online template mining library, a local sentence-embedding model, a relational store, a search index and a container runtime.

3. External Interface Requirements

3.1 User Interfaces

ID	Requirement	Pri.
IR-1.1	The system shall provide a web-based review console accessible over HTTPS from a standard browser.	High
IR-1.2	The console shall provide a source onboarding wizard covering sample submission, discovered structure, proposed mapping and approval.	High
IR-1.3	The console shall present raw and normalized representations side by side with visual linkage between corresponding fields.	High
IR-1.4	The console shall display a confidence value for every proposed field mapping and visually distinguish auto-accepted, review-pending and human-approved mappings.	High
IR-1.5	The console shall provide a review queue listing all mappings pending human adjudication, sortable by source and confidence.	High
IR-1.6	The console shall provide an operational dashboard showing ingestion rate, normalization rate, per-source volumes, confidence distribution and error counts.	Medium
IR-1.7	The console shall be usable at 1366×768 and above and shall not require any plug-in.	Medium

3.2 Software Interfaces

ID	Requirement	Pri.
IR-2.1	The system shall expose a versioned REST API over HTTPS for event query, provenance retrieval, source management and mapping approval.	High
IR-2.2	The API shall publish a machine-readable specification (OpenAPI) served by the system itself.	High
IR-2.3	The system shall interface with a relational database for mappings, schema versions, provenance and configuration.	High
IR-2.4	The system shall interface with a search index for normalized event exploration.	Medium
IR-2.5	The system shall be able to consume from and produce to a message broker topic.	Medium
IR-2.6	The system shall export normalized events as JSON, NDJSON and Parquet.	High
IR-2.7	The system shall be able to forward normalized events to an external SIEM over Syslog or HTTP.	Medium

3.3 Communications Interfaces

ID	Requirement	Pri.
IR-3.1	The system shall accept Syslog over UDP on a configurable port, handling datagram-per-event semantics.	High
IR-3.2	The system shall accept Syslog over TCP on a configurable port, handling both newline-delimited and octet-counted framing.	High
IR-3.3	The system shall accept events over HTTP and HTTPS POST, supporting single-event and batched-array payloads.	High
IR-3.4	All externally exposed HTTP interfaces shall support TLS with configurable certificates.	High
IR-3.5	The system shall not initiate any outbound connection to a public network endpoint during normal operation.	High

3.4 Hardware Interfaces

ULPF has no direct hardware interface requirements. It interacts with host resources through the container runtime: network sockets for ingestion, block storage for the vault and databases, and CPU for processing and local inference.

4. Functional Requirements

Requirements are grouped by subsystem. The Source column records the origin of each requirement: PS denotes derivation from the problem statement SIH26156, PRD denotes derivation from a feature in the Product Requirements Document, and DER denotes a requirement derived from engineering analysis.

4.1 FR-1 — Ingestion Gateway

ID	Requirement	Source	Pri.
FR-1.1	The system shall ingest log events from files, including uploaded files and files appearing in a watched directory, and shall handle rotated and partially written files without duplicating or losing events.	PS	High
FR-1.2	The system shall ingest log events from TCP and UDP network listeners on configurable ports.	PS	High
FR-1.3	The system shall ingest log events submitted to an authenticated HTTP/HTTPS endpoint, accepting single events and batches.	PS	High
FR-1.4	The system should ingest log events by consuming from a message broker topic.	PRD	Medium
FR-1.5	The system shall associate every ingested event with a registered source identity comprising a source ID, name, transport and optional vendor and product labels.	DER	High
FR-1.6	The system shall record, for every ingested event, the receipt timestamp in UTC, the transport used, the peer address where applicable, and the byte length of the original payload.	DER	High
FR-1.7	The system shall apply backpressure or bounded buffering when downstream processing cannot keep pace, and shall never silently discard an accepted event.	DER	High
FR-1.8	The system shall detect the character encoding of incoming text and shall normalize it to UTF-8 for processing while preserving the original bytes in the vault.	DER	Medium
FR-1.9	The system shall support multi-line event assembly using configurable event-boundary rules for sources that emit events spanning several lines.	DER	Medium
FR-1.10	The system shall allow ingestion to be enabled, disabled and rate-limited per source without restarting the service.	DER	Medium

4.2 FR-2 — Format Detection

ID	Requirement	Source	Pri.
FR-2.1	The system shall classify an incoming event into one of the recognised structural families: JSON, NDJSON, XML, CSV/TSV, CEF, LEEF, RFC 3164 Syslog, RFC 5424 Syslog, key-value text, delimited text, or unrecognised.	PS	High
FR-2.2	The system shall produce a numeric detection confidence for each classification decision.	DER	High
FR-2.3	The system shall route events whose detection confidence falls below a configurable threshold to structure discovery rather than rejecting them or forcing a parser.	PS	High
FR-2.4	The system shall recognise when an event originates from a source with an approved mapping and a matching template, and shall in that case bypass discovery and semantic mapping entirely.	DER	High
FR-2.5	The system shall detect when events from a known source cease to match the approved template, and shall record and surface this divergence as a format-drift indication.	DER	Medium

ID	Requirement	Source	Pri.
FR-2.6	The system shall infer the delimiter and field count for delimited text formats where a header is absent.	DER	Medium
FR-2.7	The system shall never reject an event solely because its format is unrecognised; unrecognised events shall proceed to discovery with the raw event preserved.	PS	High

4.3 FR-3 — Adaptive Parser Factory

ID	Requirement	Source	Pri.
FR-3.1	The system shall perform online template mining over the event stream of a source to discover recurring structural templates and identify constant and variable positions.	PS	High
FR-3.2	The system shall extract the variable parts of a matched template as candidate fields, each with a positional or key-based identifier.	PS	High
FR-3.3	The system shall infer a data type and semantic category for each candidate field by value-pattern analysis, covering at minimum IPv4 address, IPv6 address, port number, timestamp, MAC address, hostname, URL, file path, username, numeric quantity, boolean, and free text.	DER	High
FR-3.4	The system shall propose a target field in the unified event model for each candidate field, using deterministic rules where the source field name or position is recognised, and semantic similarity over field names and context where it is not.	PS	High
FR-3.5	The system shall compute a composite confidence score for each proposed mapping, derived from at minimum field-name similarity, value-pattern agreement with the target field's expected type, contextual consistency within the template, and historical mappings for similar fields.	DER	High
FR-3.6	The system shall automatically accept a proposed mapping only when its confidence equals or exceeds a configurable auto-accept threshold; mappings below that threshold shall be queued for human review.	PS	High
FR-3.7	The system shall persist approved mappings in a mapping registry keyed by source and template, so that subsequent events from the same source are processed deterministically without re-invoking discovery or semantic mapping.	PS	High
FR-3.8	The system shall record, for every mapping, whether it was automatically accepted or human-approved, together with the confidence score at the time of the decision.	DER	High
FR-3.9	The system should generate a deterministic, inspectable parser artefact from an approved mapping, so that the resulting extraction logic can be reviewed and exported.	PRD	Medium
FR-3.10	The system shall version mappings; changing an approved mapping shall create a new version rather than overwriting the previous one.	DER	High
FR-3.11	The system shall not use a semantic model where a deterministic rule applies; deterministic extraction shall take precedence.	DER	High

4.4 FR-4 — Normalization Engine

ID	Requirement	Source	Pri.
FR-4.1	The system shall map parsed fields into a unified event model consisting of a versioned core schema with defined field names, data types and semantics.	PS	High
FR-4.2	The core schema shall align with the Open Cybersecurity Schema Framework in its category, class, object and attribute structure.	PS	High

ID	Requirement	Source	Pri.
FR-4.3	The system should provide an export profile rendering the same normalized event using Elastic Common Schema field names.	PS	Medium
FR-4.4	The system shall normalize timestamp values to UTC in ISO-8601 format, retaining the original timestamp string and detected source timezone.	DER	High
FR-4.5	The system shall normalize action and outcome values to a controlled vocabulary, mapping vendor terms such as permit, allow, accept and pass to a single canonical value.	DER	High
FR-4.6	The system shall normalize severity values from vendor-specific scales to a single common severity scale, retaining the original value.	DER	Medium
FR-4.7	The system shall place every parsed field that does not map to a core schema field into a namespaced extensions object within the normalized event, and shall not discard it.	PS	High
FR-4.8	The system shall guarantee that the number of fields present in the parsed event equals the number of fields present in the normalized event across core and extension fields combined.	DER	High
FR-4.9	The system shall record, within each normalized event, the core schema version and the mapping version under which it was produced.	DER	High
FR-4.10	The system shall emit a normalized event even when only a subset of fields could be mapped, provided the raw event is preserved and unmapped fields are retained as extensions.	DER	High
FR-4.11	The system shall capture events that fail normalization outright into a dead-letter store together with diagnostic detail, and shall never silently discard them.	DER	High

4.5 FR-5 — Raw Event Preservation

ID	Requirement	Source	Pri.
FR-5.1	The system shall write the complete original event, byte-for-byte as received, into the Raw Event Vault before any parsing or transformation is applied.	PS	High
FR-5.2	The Raw Event Vault shall be write-once; no component shall modify or overwrite a stored raw event.	DER	High
FR-5.3	The system shall compute and store a cryptographic digest of each raw event at the time of storage.	DER	High
FR-5.4	The system shall verify the stored digest on retrieval and shall report any mismatch as an integrity failure.	DER	High
FR-5.5	The system shall store, alongside each raw event, the source identity, transport, peer address, receipt timestamp and original byte length.	DER	High
FR-5.6	The system shall provide retrieval of any raw event by its trace identifier, returning content identical to what was received.	PS	High
FR-5.7	The system shall support a configurable retention window for the Raw Event Vault, with retention expiry recorded in the audit trail.	PRD	Medium
FR-5.8	The system shall provide a verification utility that recovers every raw event in a defined corpus and reports the recoverability ratio.	DER	High
FR-5.9	The system should support reprocessing of stored raw events under a newer schema or mapping version without re-ingestion.	PRD	Medium

4.6 FR-6 — Traceability and Provenance

ID	Requirement	Source	Pri.
FR-6.1	The system shall allocate a unique trace identifier to every ingested event at the moment of ingestion, before any processing.	PS	High
FR-6.2	The trace identifier shall be carried on the raw, parsed and normalized representations of the event.	PS	High
FR-6.3	The system shall resolve a normalized event to its originating raw event using the trace identifier.	PS	High
FR-6.4	The system shall resolve a raw event forward to the normalized event or events derived from it.	PS	High
FR-6.5	The system shall record, for each field in a normalized event, the source field name or positional index from which it was derived.	PS	High
FR-6.6	The system shall record, for each normalized field, the transformation applied, the mapping identifier and version used, and the confidence score assigned at mapping time.	DER	High
FR-6.7	The system shall expose an API returning the complete derivation chain for a given event or a given field within that event.	PS	High
FR-6.8	The system shall retain provenance records for at least as long as the corresponding normalized events are retained.	DER	High
FR-6.9	The system shall record an immutable audit entry for every mapping approval, correction and schema version change, including actor identity and timestamp.	DER	High

4.7 FR-7 — Schema Registry and Evolution

ID	Requirement	Source	Pri.
FR-7.1	The system shall maintain a schema registry holding every version of the core schema, each identified by an immutable version identifier.	PS	High
FR-7.2	A published schema version shall be immutable; any change shall produce a new version.	DER	High
FR-7.3	The system shall permit only additive changes within a major schema version; field removal or type change shall require a new major version.	DER	High
FR-7.4	The system shall support vendor and source extension namespaces that are isolated from the core schema and cannot collide with core field names.	PS	High
FR-7.5	The system shall allow a schema version to be pinned per source, so that a schema upgrade does not implicitly change the output of an existing source.	DER	Medium
FR-7.6	The system shall validate every normalized event against the declared schema version before emission and shall route validation failures to the dead-letter store.	DER	High
FR-7.7	The system shall accept an event containing a previously unseen field without failure, preserving that field as an extension.	PS	High
FR-7.8	The system should support export and import of schema versions and mapping packages as files, enabling transfer between air-gapped deployments.	PRD	Medium

4.8 FR-8 — Review Console and Human-in-the-Loop

ID	Requirement	Source	Pri.
FR-8.1	The system shall provide a guided onboarding workflow allowing a user to submit a sample from a new source and receive the discovered structure and proposed mapping.	PS	High

ID	Requirement	Source	Pri.
FR-8.2	The system shall maintain a review queue containing every mapping proposal whose confidence fell below the auto-accept threshold.	PS	High
FR-8.3	The system shall present, for each proposal under review, the source field, sample values, inferred type, proposed target field and the constituent signals contributing to the confidence score.	DER	High
FR-8.4	The system shall allow a reviewer to accept a proposal, reassign a field to a different target, mark a field as extension-only, or reject the mapping.	PS	High
FR-8.5	The system shall persist reviewer decisions as a new mapping version and shall apply that version to subsequent events from the source.	DER	High
FR-8.6	The system shall continue to ingest, preserve and trace events from a source while its mapping is pending review; such events shall be normalized on a best-effort basis with unmapped fields retained as extensions.	DER	High
FR-8.7	The system shall provide a side-by-side inspector displaying a raw event and its normalized form with per-field linkage.	PRD	High
FR-8.8	The system should provide an operational dashboard reporting ingestion and normalization rates, per-source volumes, confidence distribution, review-queue depth and error counts.	PRD	Medium
FR-8.9	The system shall record the identity of the approving user and the timestamp for every mapping decision.	DER	High

4.9 FR-9 — Downstream Delivery

ID	Requirement	Source	Pri.
FR-9.1	The system shall expose a REST API for querying normalized events with filtering by source, time range, event category and field values, with pagination.	PS	High
FR-9.2	The query API shall optionally expand provenance and include a reference to the raw event.	DER	High
FR-9.3	The system shall export normalized events as JSON, NDJSON and Parquet files to a configurable output location.	PS	High
FR-9.4	The system should index normalized events into a search engine to support interactive exploration.	PRD	Medium
FR-9.5	The system should forward normalized events to an external SIEM or collector over Syslog, HTTP or a message broker.	PS	Medium
FR-9.6	The system shall make dead-lettered events retrievable with their failure diagnostics and their raw event reference.	DER	High
FR-9.7	The system shall ensure that a normalized event delivered downstream always carries its trace identifier, schema version and mapping version.	DER	High

4.10 FR-10 — Deployment, Configuration and Administration

ID	Requirement	Source	Pri.
FR-10.1	The system shall be deployable as a set of containers started by a single Compose command.	PS	High
FR-10.2	The system shall install and operate with no outbound network access available at any stage, including model weights bundled at build time.	PS	High
FR-10.3	The system shall express all sources, transports, thresholds, schema profiles and output sinks in file-based configuration suitable for version control.	DER	High

ID	Requirement	Source	Pri.
FR-10.4	The system shall expose liveness, readiness and metrics endpoints for operational monitoring.	PRD	Medium
FR-10.5	The system should enforce role-based access control distinguishing viewer, approver and administrator roles.	PRD	Medium
FR-10.6	The system should support running multiple stateless processing workers consuming from a partitioned queue, scaled without code change.	PS	Medium
FR-10.7	The system shall write structured application logs of its own operation, separate from the event data it processes.	DER	High
FR-10.8	The system shall support configuration reload for source definitions and thresholds without a full restart.	DER	Low

5. Data Requirements

5.1 Principal Data Entities

ID	Entity	Description	Key attributes
DR-1	Raw Event	The original event exactly as received	trace_id, source_id, received_at, transport, peer, byte_length, digest, payload
DR-2	Source	A registered origin of events	source_id, name, transport, vendor, product, schema_pin, status
DR-3	Template	A discovered structural pattern for a source	template_id, source_id, pattern, variable_positions, first_seen, occurrence_count
DR-4	Candidate Field	A variable part extracted from a template	field_key, position, inferred_type, sample_values
DR-5	Mapping	The approved association of source fields to target fields	mapping_id, source_id, template_id, version, field_bindings, status, approved_by, approved_at
DR-6	Normalized Event	The output event in the unified event model	trace_id, schema_version, mapping_version, core fields, extensions
DR-7	Provenance Record	Per-field derivation of a normalized event	trace_id, target_field, source_field, transformation, mapping_version, confidence
DR-8	Schema Version	An immutable version of the core schema	schema_version, published_at, field_definitions, compatibility_class
DR-9	Audit Entry	An immutable record of a governance action	audit_id, actor, action, subject, before, after, timestamp
DR-10	Dead Letter	An event that failed processing	trace_id, stage, error_class, diagnostic, raw_reference, occurred_at

5.2 Unified Event Model — Core Field Groups

The core schema is organised into object groups aligned with OCSF. The table below defines the minimum core surface required for the prototype; the full field catalogue is maintained in the schema registry.

Group	Representative fields	Purpose
metadata	trace_id, schema_version, mapping_version, product.vendor, product.name	Identity and lineage of the normalized record
time	event.time_utc, event.time_original, event.timezone, received_at	Normalized and original temporal information
source	source.ip, source.port, source.hostname, source.user, source.mac	The originating endpoint of the observed activity
destination	destination.ip, destination.port, destination.hostname	The target endpoint of the observed activity
network	network.protocol, network.direction, network.bytes, network.packets	Transport-level characteristics
event	event.category, event.class, event.action, event.outcome, event.severity	Classification and outcome of the activity
device	device.hostname, device.ip, device.type	The device that reported the event
observer	observer.source_id, observer.transport, observer.peer	How ULPF received the event

Group	Representative fields	Purpose
extensions	extensions.<namespace>.<field>	Source-specific fields preserved without loss
raw_ref	raw_ref.trace_id, raw_ref.digest, raw_ref.byte_length	Pointer and integrity reference to the preserved original

5.3 Illustrative Normalized Event

The example below shows a previously unseen vendor record and the normalized output, including the preservation of a vendor field that has no core equivalent.

INPUT (unseen proprietary format, received over UDP Syslog)
2026-08-22 18:41 FW-X 10.2.1.5 -> 10.5.4.8 TCP/22 DENY RID=82931

OUTPUT (unified event model, abbreviated)

```
{
  "metadata": {
    "trace_id": "ULPF-8F91-4C2A-B7D0",
    "schema_version": "ulpf-core-1.2",
    "mapping_version": "map-fwx-3"
  },
  "time": {
    "event.time_utc": "2026-08-22T13:11:00Z",
    "event.time_original": "2026-08-22 18:41"
  },
  "source": {
    "ip": "10.2.1.5"
  },
  "destination": {
    "ip": "10.5.4.8", "port": 22
  },
  "network": {
    "protocol": "tcp"
  },
  "event": {
    "category": "network_activity", "action": "deny"
  },
  "device": {
    "hostname": "FW-X"
  },
  "extensions": {
    "fwx": { "RID": "82931" }
  },
  "raw_ref": {
    "trace_id": "ULPF-8F91-4C2A-B7D0",
    "digest": "sha256:9c1e...", "byte_length": 58
  }
}
```

5.4 Data Integrity Requirements

- DR-I1 — A raw event, once written, shall never be altered by any process.
- DR-I2 — Every normalized event shall reference exactly one raw event by trace identifier.
- DR-I3 — A provenance record shall not exist without a corresponding normalized event and target field.
- DR-I4 — A mapping version referenced by a normalized event shall remain retrievable for as long as that event is retained.
- DR-I5 — Audit entries shall be append-only.

5.5 Data Retention

Data class	Default retention	Configurable
Raw events	90 days	Yes — per source
Normalized events	90 days in the index; exported files retained by the data platform	Yes
Provenance records	Equal to normalized event retention	Bound to events
Mappings and templates	Indefinite	No — required for reprocessing
Schema versions	Indefinite	No
Audit entries	Indefinite	No
Dead letters	30 days	Yes

6. Non-Functional Requirements

6.1 Performance

ID	Requirement	Pri.
NFR-P1	The system shall process events from a source with an approved mapping on a deterministic fast path that does not invoke template mining or semantic inference.	High
NFR-P2	The system shall achieve a sustained normalization throughput measured and reported on defined reference hardware and a defined workload; the measured figure shall be published rather than estimated.	High
NFR-P3	The p95 normalization latency on the fast path shall be under 100 ms per event on reference hardware.	High
NFR-P4	Structure discovery for a new source shall return a proposed mapping for an interactive sample within 60 seconds.	High
NFR-P5	The complete onboarding workflow for an unseen source, from first log to validated normalized events, shall complete within 10 minutes for the benchmark scenario.	High
NFR-P6	Semantic mapping inference shall execute on CPU without requiring a hardware accelerator.	High
NFR-P7	Query API responses for a bounded result page shall return within 2 seconds under nominal load.	Medium

6.2 Scalability

ID	Requirement	Pri.
NFR-S1	Processing workers shall be stateless with respect to event content, so that additional workers can be added without coordination.	High
NFR-S2	The event stream shall be partitionable by source so that ordering is preserved within a source while parallelism is achieved across sources.	Medium
NFR-S3	Adding a new log source shall not require redeployment or restart of the processing tier.	High
NFR-S4	Storage growth shall be bounded by configurable retention rather than by unbounded accumulation.	Medium
NFR-S5	The system shall document its measured single-node capacity and the architectural path to multi-node scale, without claiming unmeasured throughput.	High

6.3 Reliability and Availability

ID	Requirement	Pri.
NFR-R1	No accepted event shall be lost; an event that cannot be processed shall appear in the dead-letter store with diagnostics.	High
NFR-R2	The system shall persist the raw event before acknowledging ingestion, so that a crash after acknowledgement cannot lose data.	High
NFR-R3	Processing shall be idempotent with respect to trace identifier, so that a retried event does not create duplicate normalized records.	High
NFR-R4	A failure in semantic mapping shall degrade to extension-only preservation rather than failing the event.	High
NFR-R5	The system shall recover to a consistent state after an unclean shutdown without manual intervention.	Medium

ID	Requirement	Pri.
NFR-R6	Target availability for the ingestion tier shall be 99.5% in a single-node prototype deployment.	Medium

6.4 Security

ID	Requirement	Pri.
NFR-SE1	All externally exposed HTTP interfaces shall require TLS.	High
NFR-SE2	API access shall require authentication; anonymous access shall not be permitted to event or provenance data.	High
NFR-SE3	The system shall enforce role-based authorisation distinguishing read, approve and administer privileges.	Medium
NFR-SE4	The system shall not transmit log content to any external service under any configuration.	High
NFR-SE5	The system shall treat all ingested content as untrusted input and shall be resistant to injection through log content, including parser-directed and query-directed injection.	High
NFR-SE6	Credentials and secrets shall be supplied through environment or secret files, never embedded in images or configuration committed to version control.	High
NFR-SE7	The system should support optional field-level masking or redaction for configured sensitive fields, applied to the normalized event while the raw event remains protected under access control.	Medium
NFR-SE8	All governance actions shall be recorded in an append-only audit trail.	High

6.5 Usability

ID	Requirement	Pri.
NFR-U1	An engineer familiar with log formats but unfamiliar with ULPF shall be able to complete a source onboarding without reading documentation beyond in-product guidance.	High
NFR-U2	Confidence values shall be presented with the evidence that produced them, not as an unexplained number.	High
NFR-U3	The console shall make it visually unambiguous which fields were auto-accepted, which are pending review, and which were human-corrected.	High
NFR-U4	Error messages shall state what failed, at which pipeline stage, and what the operator can do about it.	Medium

6.6 Maintainability and Portability

ID	Requirement	Pri.
NFR-M1	Each subsystem shall be independently testable with defined inputs and outputs.	High
NFR-M2	Adding support for a new deterministic format shall require adding a parser module and registering it, with no change to the pipeline core.	High
NFR-M3	Adding a new target schema profile shall require adding a mapping profile, with no change to the normalization engine.	Medium
NFR-M4	The system shall run on any host providing a compatible container runtime, without OS-specific code paths.	High
NFR-M5	All third-party dependencies shall be pinned to explicit versions and vendored into the offline bundle.	High

ID	Requirement	Pri.
NFR-M6	The system shall achieve at least 70% automated test coverage over the parsing, mapping and normalization modules.	Medium

6.7 Compliance and Auditability

ID	Requirement	Pri.
NFR-C1	For any normalized event presented as evidence, the system shall be able to produce the original record and the transformation chain that produced it.	High
NFR-C2	The integrity of a preserved raw event shall be demonstrable through a stored and re-verifiable digest.	High
NFR-C3	The system shall record which schema and mapping version was in force at the time each event was processed, so that historical output is explicable.	High
NFR-C4	The system shall operate entirely within the customer's network boundary, supporting data-residency and sovereignty requirements.	High

7. Requirements Traceability

7.1 Problem Statement Coverage

The table maps each capability named in problem statement SIH26156 to the requirements that satisfy it. This is the primary completeness check for the submission.

PS capability	Satisfied by	Verified by
Ingest logs of any format from any source	FR-1.1 to FR-1.10, FR-2.1, FR-2.7	TC-ING-01 to TC-ING-08
Automatic parsing and structure discovery	FR-2.1 to FR-2.6, FR-3.1 to FR-3.3	TC-DIS-01 to TC-DIS-06
Normalization to a common schema	FR-4.1 to FR-4.6, FR-4.9	TC-NRM-01 to TC-NRM-07
Standardization (OCSF / ECS)	FR-4.2, FR-4.3, FR-7.1	TC-SCH-01 to TC-SCH-04
100% lossless preservation of the original	FR-5.1 to FR-5.8, FR-4.7, FR-4.8	TC-PRS-01 to TC-PRS-05
Traceability between raw and processed data	FR-6.1 to FR-6.9	TC-TRC-01 to TC-TRC-06
Plug-and-play onboarding of new sources	FR-3.4 to FR-3.10, FR-8.1 to FR-8.6	TC-ONB-01 to TC-ONB-05
Interoperability with SIEM and data lakes	FR-9.1 to FR-9.7, IR-2.6, IR-2.7	TC-OUT-01 to TC-OUT-05
AI/ML readiness of output	FR-4.1, FR-4.4 to FR-4.6, FR-9.3	TC-OUT-03, TC-NRM-04
Air-gapped deployment	FR-10.2, NFR-SE4, C1, C5	TC-DEP-01 to TC-DEP-03
Containerized deployment	FR-10.1, NFR-M4	TC-DEP-01
Scalability to large event volumes	FR-10.6, NFR-S1 to NFR-S5	TC-PRF-01 to TC-PRF-04

7.2 Goal to Requirement Traceability

PRD goal	Requirements
G1 — Zero/minimal parser development	FR-3.1 to FR-3.11, FR-8.1 to FR-8.6, NFR-P4, NFR-P5
G2 — Complete recoverability of the original	FR-5.1 to FR-5.8, DR-I1, NFR-C2
G3 — Field-level provenance	FR-6.5 to FR-6.8, DR-7, NFR-C1
G4 — Open-schema aligned output	FR-4.2, FR-4.3, FR-7.1 to FR-7.4
G5 — Absorb schema and format change	FR-2.5, FR-7.2 to FR-7.7, FR-4.7
G6 — Fully offline containerised operation	FR-10.1, FR-10.2, NFR-SE4, NFR-M4, NFR-M5
G7 — Human control over uncertain decisions	FR-3.5, FR-3.6, FR-8.2 to FR-8.5, FR-8.9

7.3 Verification Methods

Method	Applied to	Evidence produced
Unit test	Parsers, type inference, value normalizers, confidence computation	Automated test report with coverage
Integration test	End-to-end pipeline from transport to normalized output	Pipeline test report per format family
Corpus test	Full test matrix of formats, structures, transports and schema conditions	Coverage ratio against the test matrix
Ground-truth comparison	Field mapping accuracy	Accuracy percentage against a labelled field set
Recoverability audit	Raw-event preservation	Recovered / ingested ratio across the full corpus
Benchmark	Throughput and latency	Measured figures with stated hardware and workload
Air-gap test	Offline installation and operation	Egress monitor log showing zero outbound connections
Inspection / review	Documentation, configuration, audit trail completeness	Review checklist sign-off

8. Validation Approach

8.1 Universality Test Matrix

Because “universal” cannot be proven by enumerating every possible format, coverage is measured across a defined matrix of dimensions. Coverage is reported as successfully processed test categories divided by total test categories.

Dimension	Test categories
Format	Syslog RFC 3164, Syslog RFC 5424, JSON, NDJSON, XML, CSV, TSV, CEF, LEEF, key-value text, delimited proprietary, positional proprietary
Source type	Firewall, router, server OS, web application, database, endpoint agent, VPN, identity provider, cloud audit trail
Structure	Fully structured, semi-structured, free text with embedded fields, multi-line
Schema knowledge	Known source with approved mapping, known format with unknown source, entirely unknown format
Field composition	Standard fields only, standard plus vendor-specific fields, vendor-specific fields only
Evolution	Field added, field renamed, field order changed, template variant introduced
Transport	File, TCP, UDP, HTTP, message broker
Adversarial input	Truncated event, malformed encoding, injected delimiter, oversized field, embedded newline

8.2 Acceptance Criteria for the Prototype

ID	Acceptance criterion
AC-1	An event of a previously unseen proprietary format is ingested, its structure discovered, its fields extracted and typed, and a mapping proposed with per-field confidence — with no source-specific code written.
AC-2	Every event ingested during the acceptance run is recoverable byte-identically from the Raw Event Vault, with digest verification passing.
AC-3	Every field present in a parsed event is present in the normalized event, either as a core field or under extensions.
AC-4	For any field of any normalized event, the provenance API returns the source field, transformation, mapping version and confidence.
AC-5	A mapping proposal scoring below the auto-accept threshold is held in the review queue and is not applied until a reviewer approves it.
AC-6	The full stack installs and operates on a host with no outbound network route, verified by an egress monitor.
AC-7	An event containing a previously unseen field is processed successfully, with the new field preserved and the divergence surfaced.
AC-8	Throughput and latency figures are reported as measured values with the hardware and workload stated.

8.3 Reporting Discipline

Any metric not yet measured is reported as a target and labelled as such. Measured results replace targets only when produced by the benchmark harness on stated hardware against the stated corpus. This applies in particular to throughput, latency and field mapping accuracy, which are the figures most likely to be challenged during evaluation.